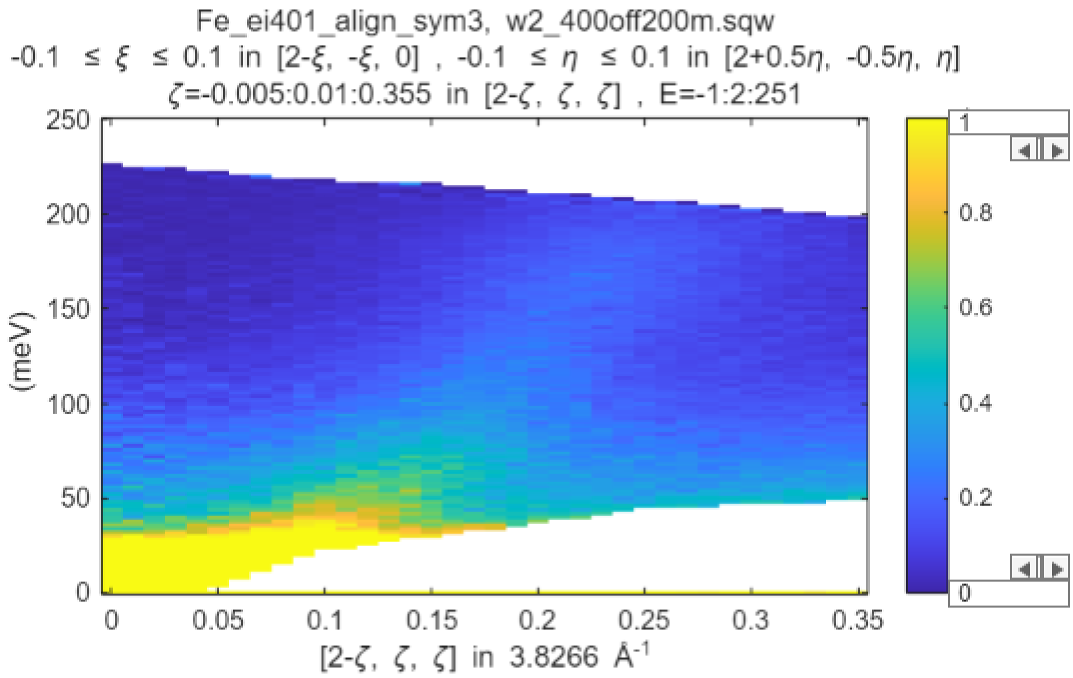
```
root_path = fileparts(mfilename("fullpath"));
root_path  = pwd;
data_path = fullfile(root_path,'sym4D_cutsAndFits');

if ~exist('cuts2fit400GP','var')
    ld = load(fullfile(data_path,'multicuts_fit_dataGP_ei400meV.mat'));
    cuts2fit400 = ld.cuts2fit400;
end
if exist('used_ds400GP','var')
    used_ds    = used_ds400GP;
else
    used_ds = containers.Map()
    sel_info = struct('cuts',[],'en_range',[],'fname','','bg_par',[]);
    %
    sel_info.cuts = {cuts2fit400GP.cuts_off110f};
    sel_info.en_range = 50:10:160;
    sel_info.cut_dim  = 2;
    sel_info.fname='EnFit_Ei400dir111off110_2D';
    sel_info.bg_par= {};
    used_ds('Ei400Br110dir111')=sel_info;
    %
    sel_info.cuts = {cuts2fit400GP.cuts_off200f};
    sel_info.en_range = 50:10:200;
    sel_info.cut_dim  = 2;
    sel_info.fname='EnFit_Ei400dir111off200_2D';
    sel_info.bg_par= {};
    used_ds('Ei400Br200dir111')=sel_info;

    used_ds400GP = used_ds;
end
```

```
used_ds =

  Map with properties:

        Count: 0
      KeyType: char
    ValueType: any

  Show all accessible properties of Map
```

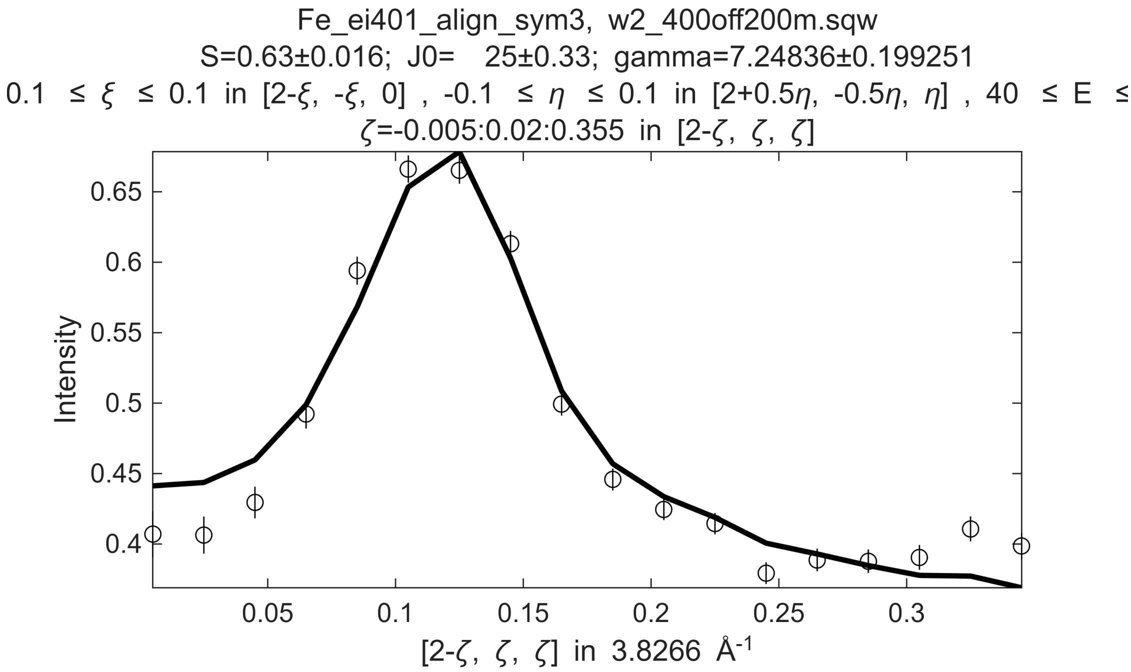
```
select= 'Ei400Br200dir111';
sel_info = used_ds(select);

cuts_list = sel_info.cuts;

mi = maps_instrument(400,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end
```

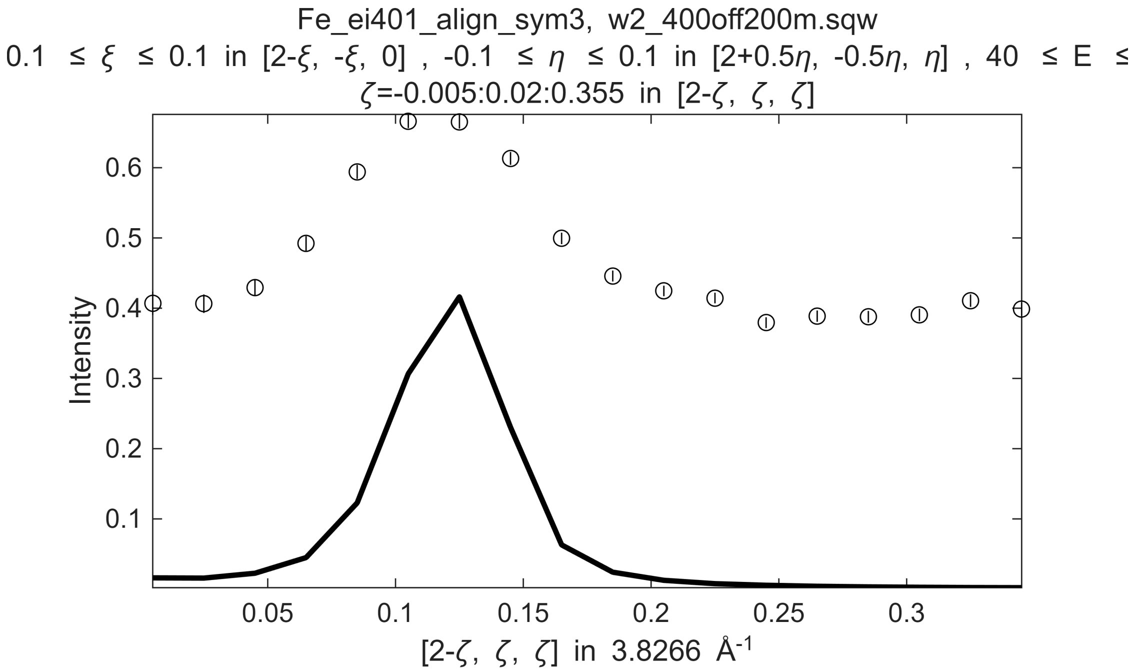



```
dir_name  = "GP";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning
cut_en = sel_info.en_range; %cuts2fit400.cuts_dir001FFfit(2);
cut_dimensions = sel_info.cut_dim;

bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name  = sel_info.fname;
```

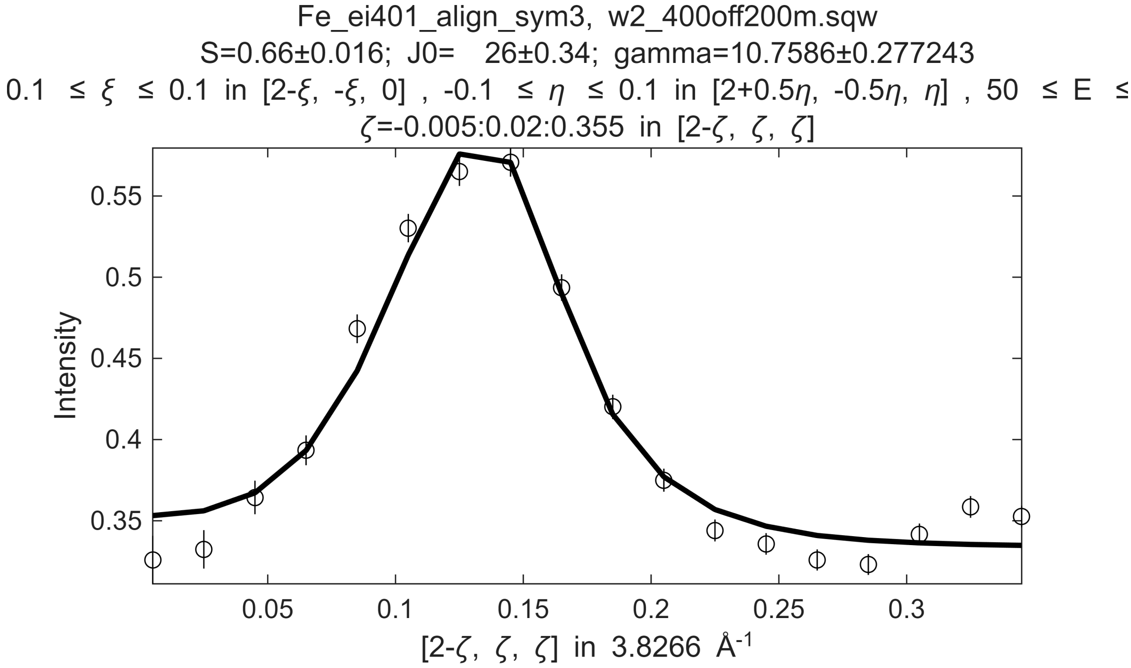
```
[fit_res_400rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);
```

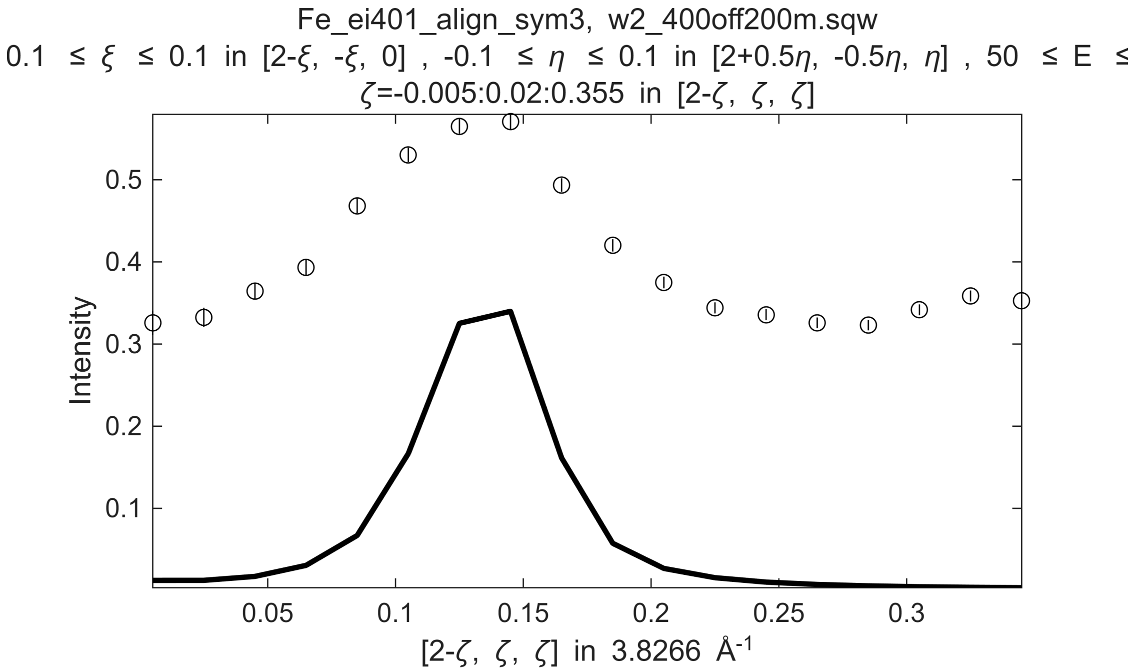
```
******************************
**** En = 50±10
******************************
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S=0.63±0.016; J0= 25±0.33; gamma=7.24836±0.199251
0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 40 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1


Fe_ei401_align_sym3, w2_400off200m.sqw
0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 40 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

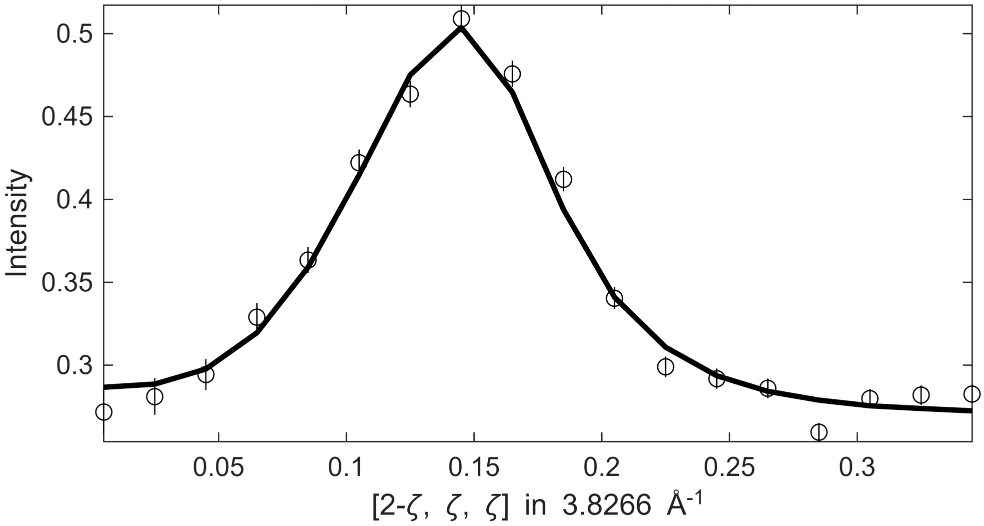
```
******************************
**** En = 60±10
******************************
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S=0.66±0.016; J0= 26±0.34; gamma=10.7586±0.277243
0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 50 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

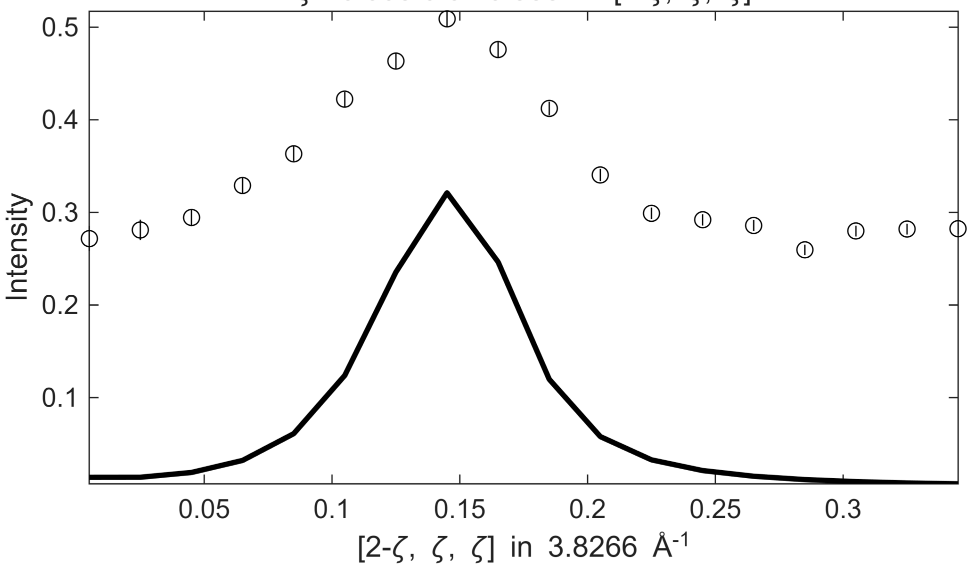

Fe_ei401_align_sym3, w2_400off200m.sqw
0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 50 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

```
******************************
**** En = 70±10
******************************
```

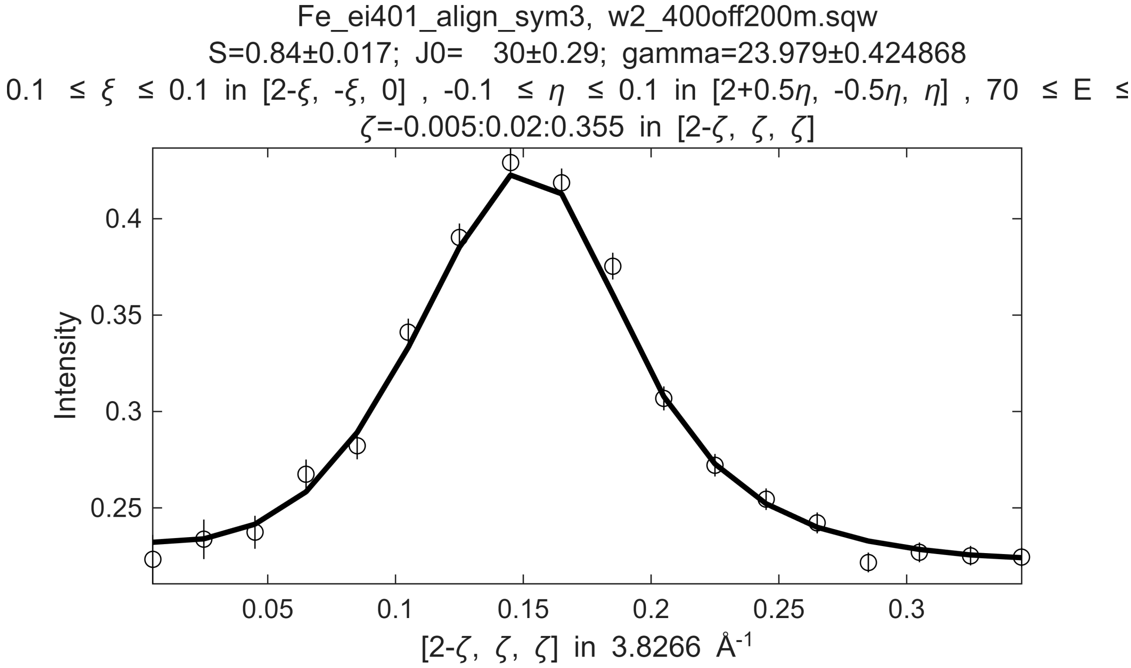
Fe_ei401_align_sym3, w2_400off200m.sqw
S= 0.8±0.016; J0= 28±0.27; gamma=18.0243±0.331781
0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 60 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]

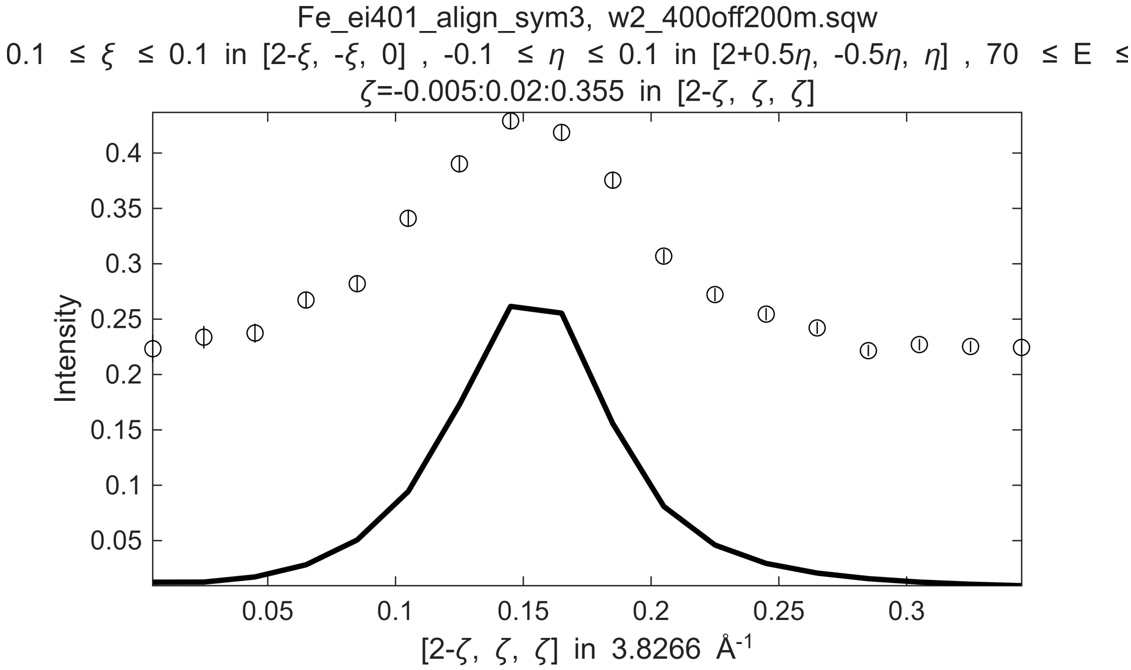


Fe_ei401_align_sym3, w2_400off200m.sqw
0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 60 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]

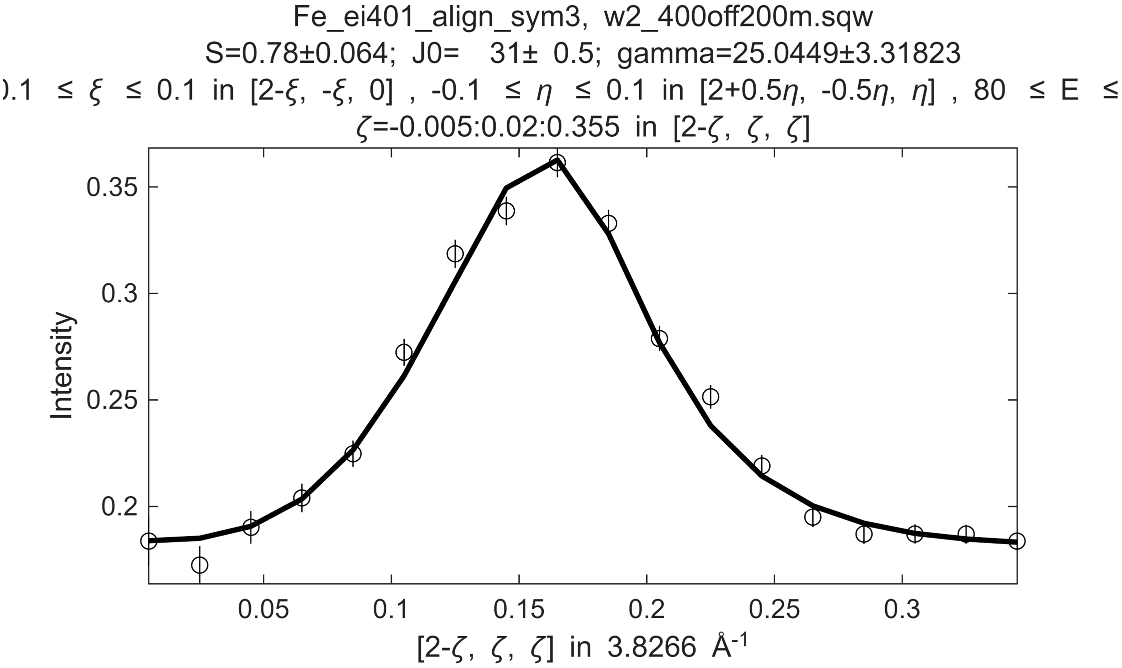


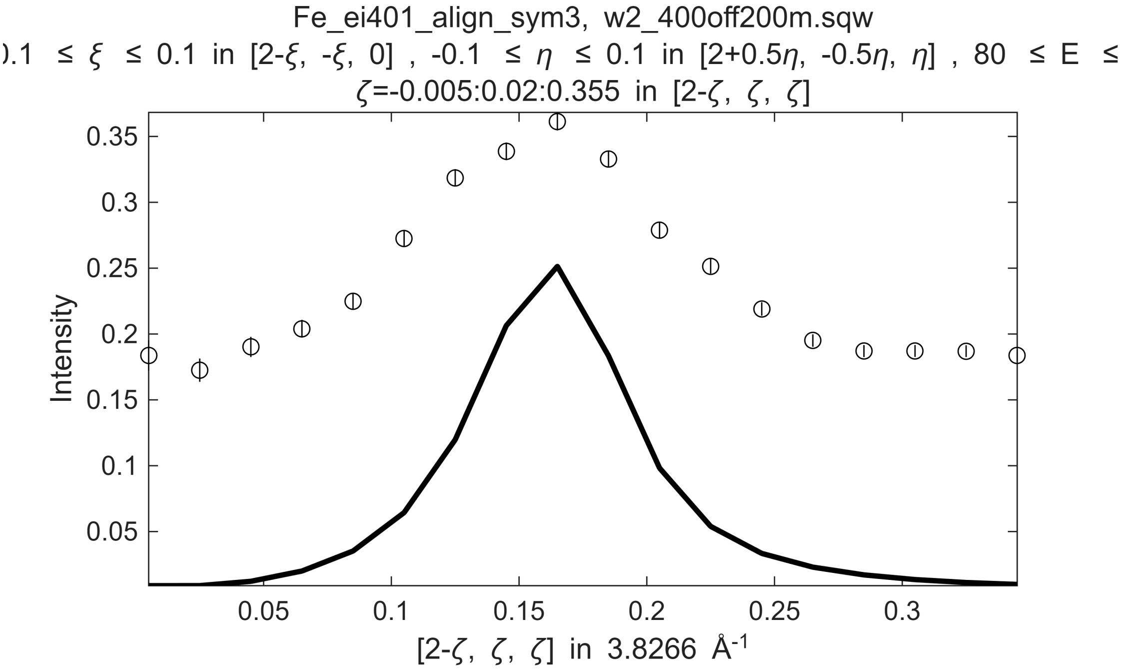

```
******************************
**** En = 80±10
******************************
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S=0.84±0.017; J0= 30±0.29; gamma=23.979±0.424868
0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 70 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

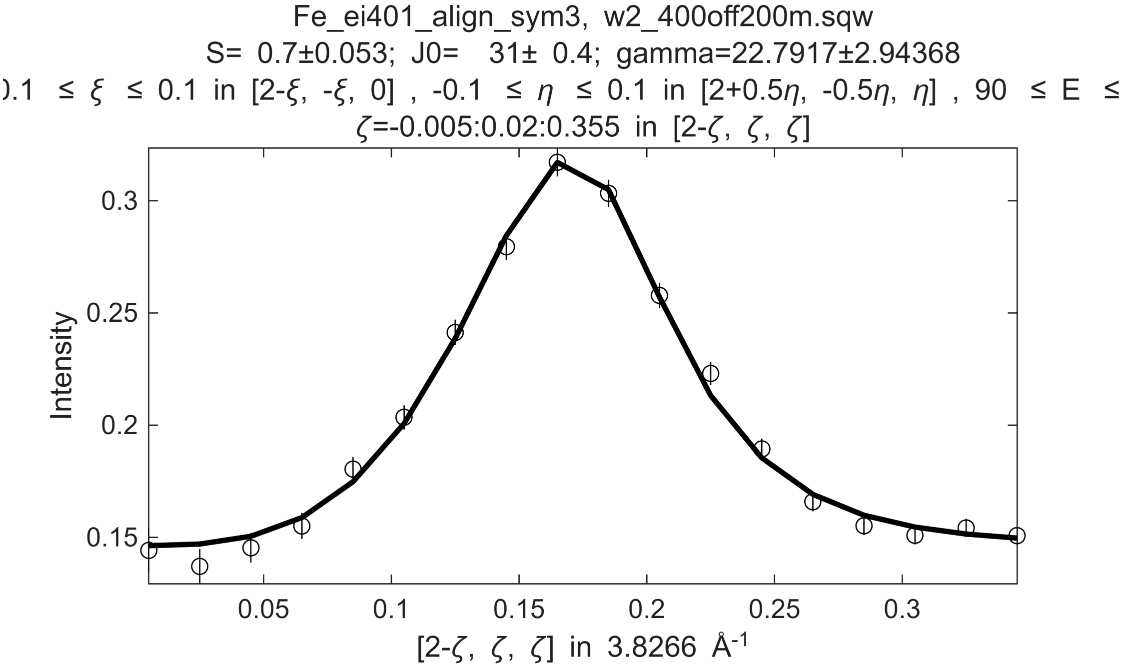

Fe_ei401_align_sym3, w2_400off200m.sqw
0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 70 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

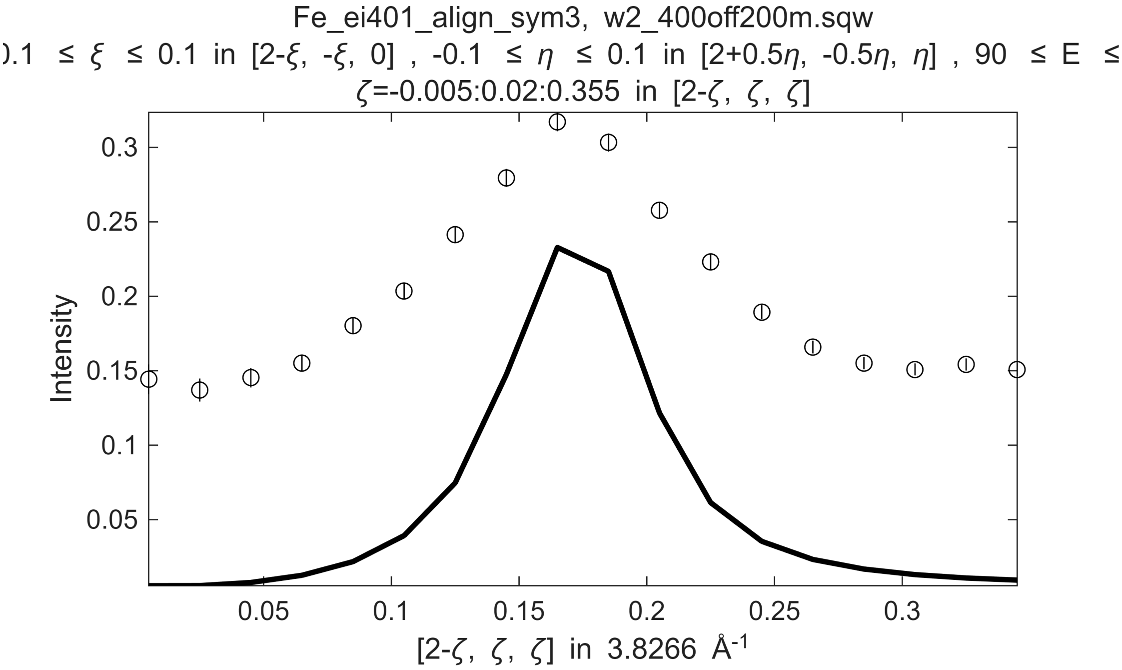
```
******************************
**** En = 90±10
******************************
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S=0.78±0.064; J0= 31± 0.5; gamma=25.0449±3.31823
).1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 80 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
0.35
0.3
0.25
0.2
0.05
0.1
0.15
0.2
0.25
0.3
[2-ζ, ζ, ζ] in 3.8266 Å-1


Fe_ei401_align_sym3, w2_400off200m.sqw
).1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 80 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0.05
0.1
0.15
0.2
0.25
0.3
[2-ζ, ζ, ζ] in 3.8266 Å-1

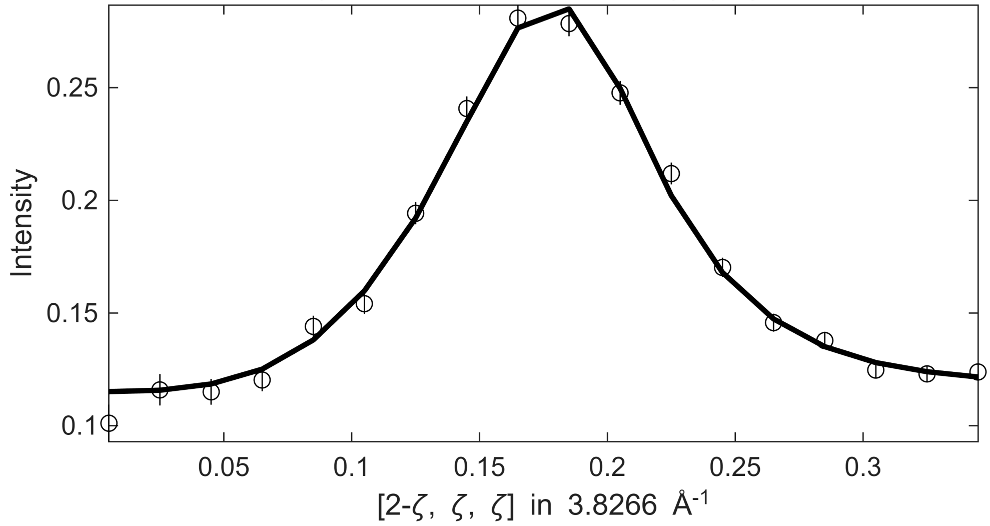
```
******************************
**** En = 100±10
******************************
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S= 0.7±0.053; J0= 31± 0.4; gamma=22.7917±2.94368
).1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 90 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

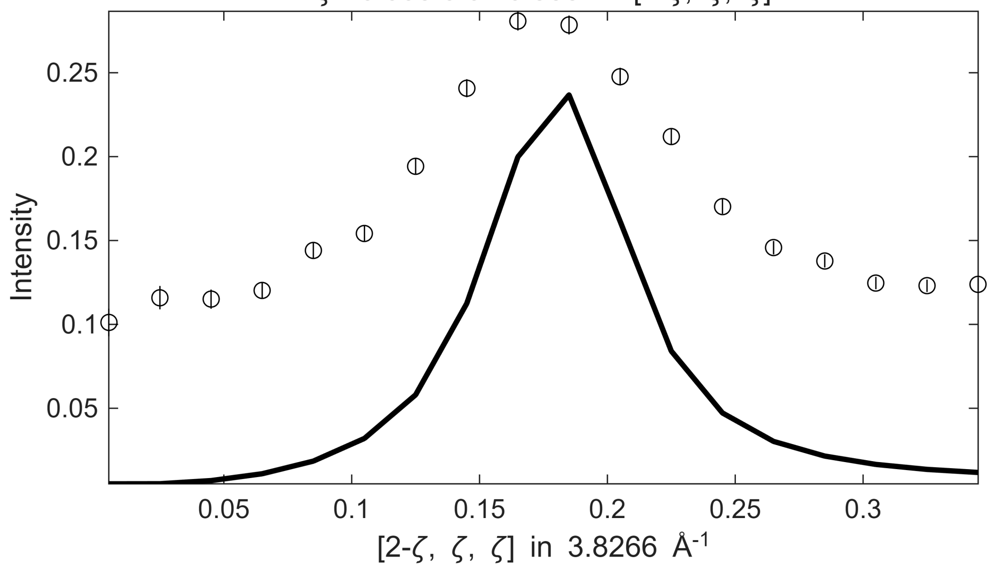

Fe_ei401_align_sym3, w2_400off200m.sqw
).1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 90 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

```
******************************
**** En = 110±10
******************************
```

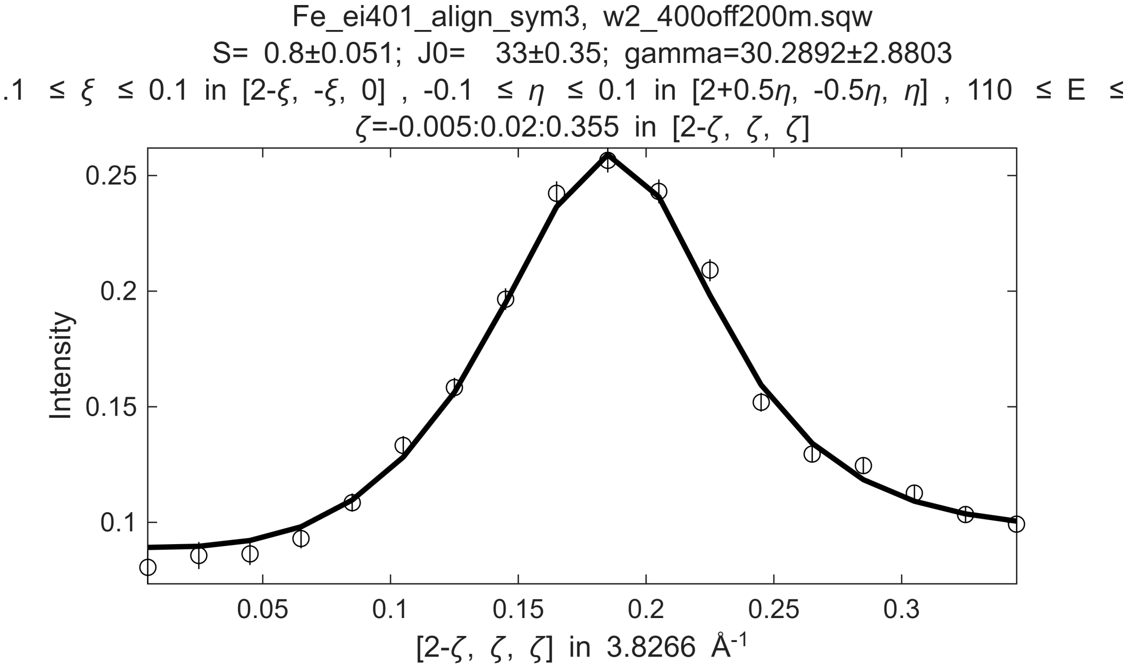
Fe_ei401_align_sym3, w2_400off200m.sqw
S=0.74±0.051; J0= 32±0.36; gamma=25.0336±2.8394
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 100 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]

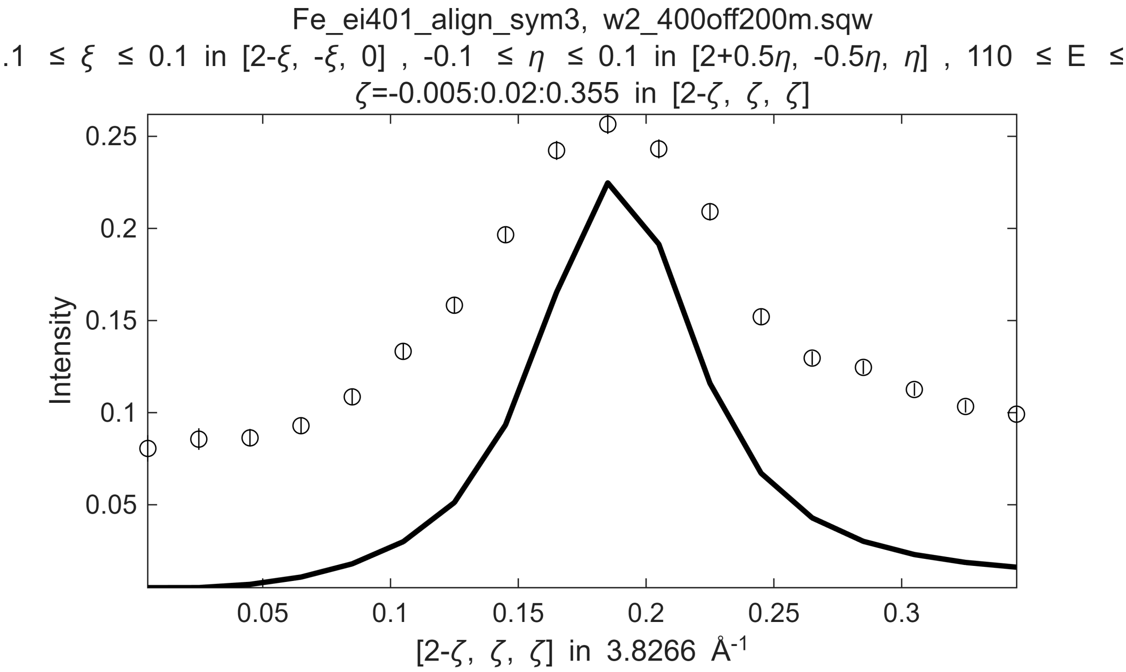


Fe_ei401_align_sym3, w2_400off200m.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 100 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]



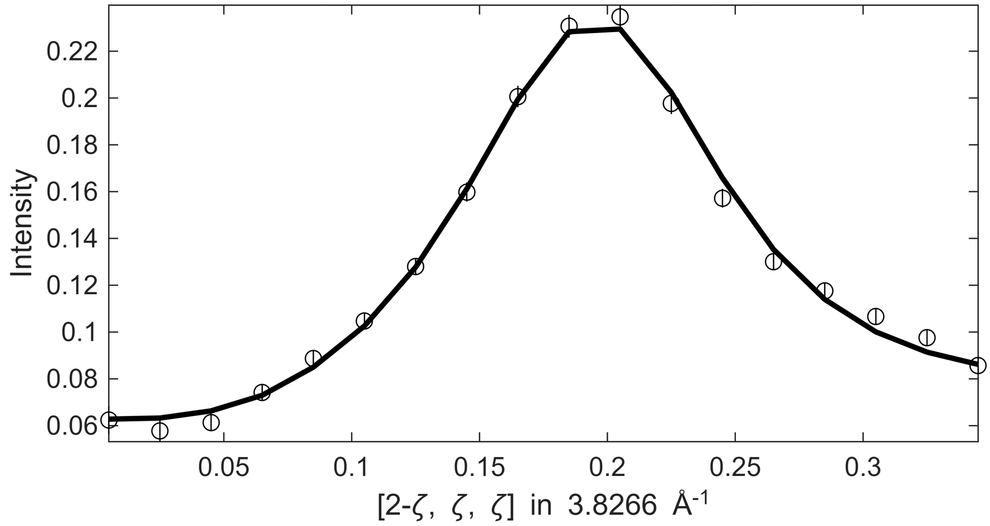

```
******************************
**** En = 120±10
******************************
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S= 0.8±0.051; J0= 33±0.35; gamma=30.2892±2.8803
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 110 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å^-1

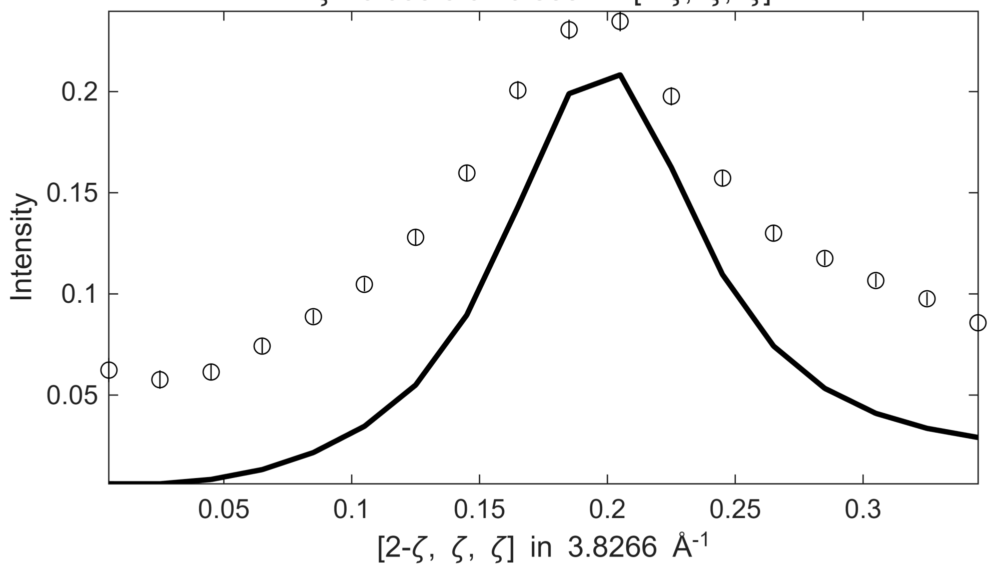

Fe_ei401_align_sym3, w2_400off200m.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 110 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å^-1

```
******************************
**** En = 130±10
******************************
```

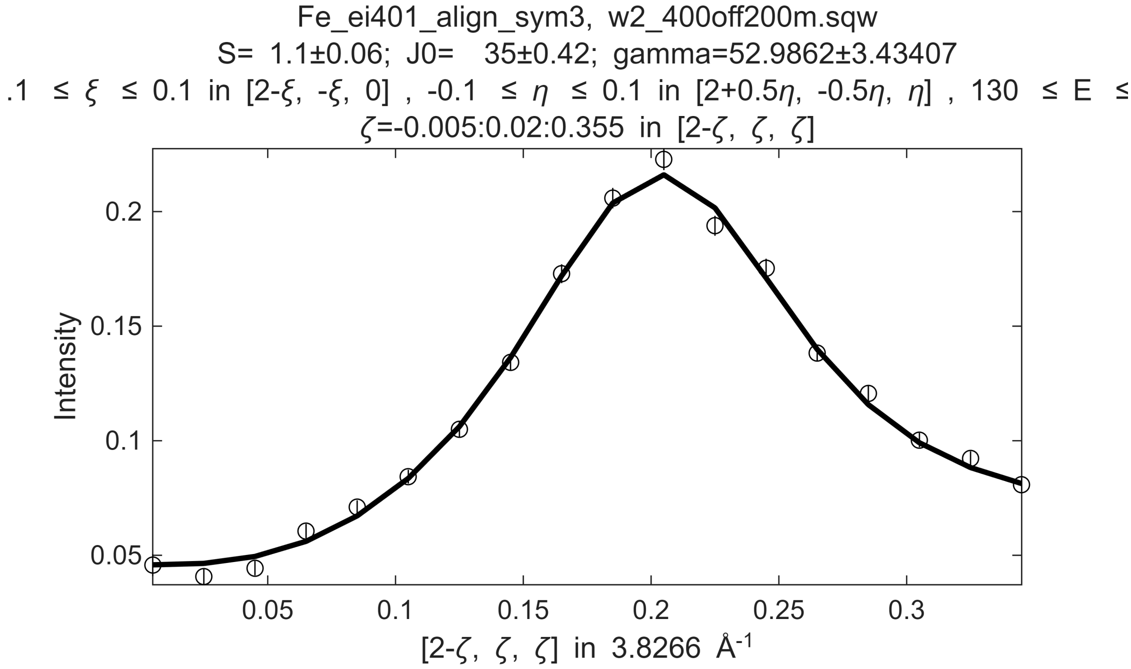
Fe_ei401_align_sym3, w2_400off200m.sqw
S= 1±0.065; J0= 34±0.44; gamma=43.2885±3.62373
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 120 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]

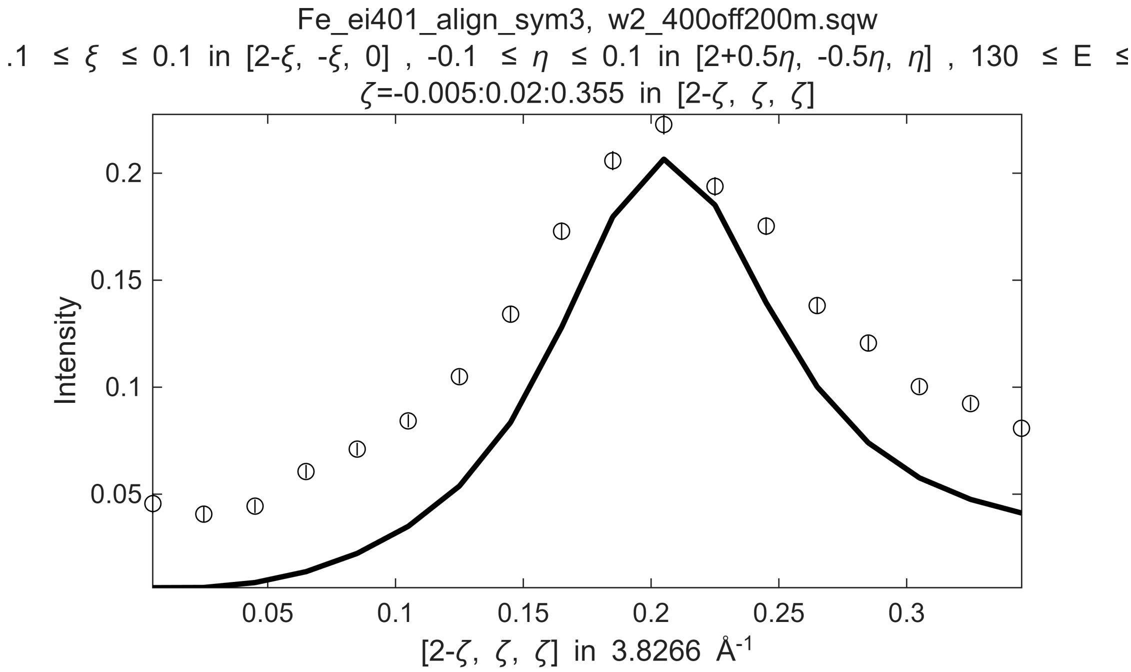


Fe_ei401_align_sym3, w2_400off200m.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 120 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]

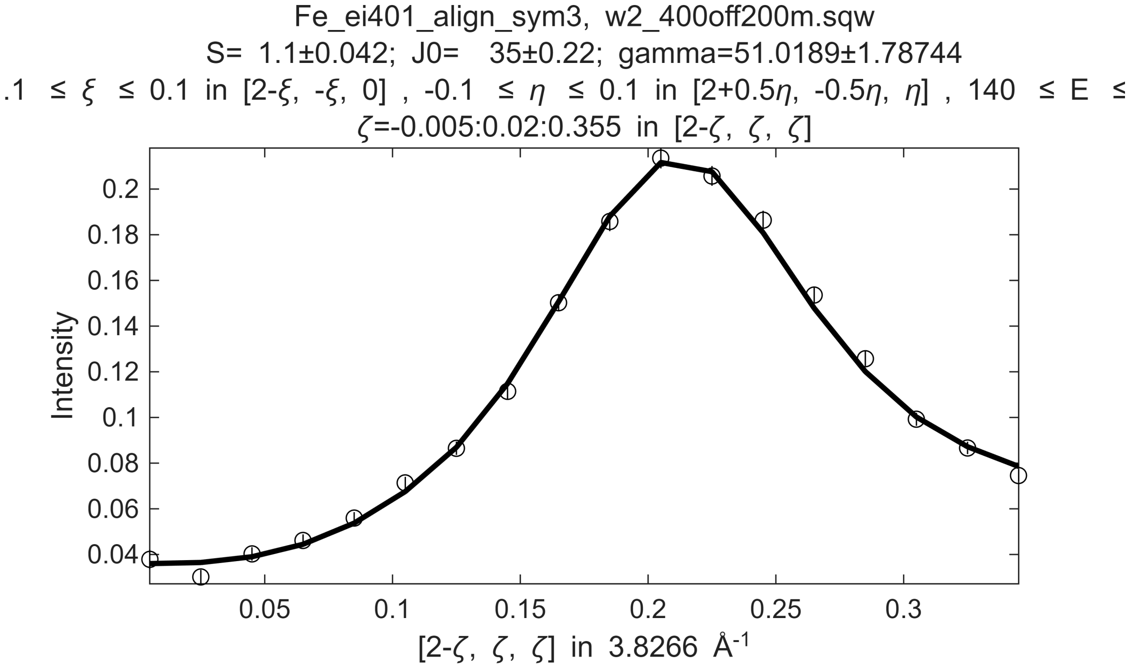


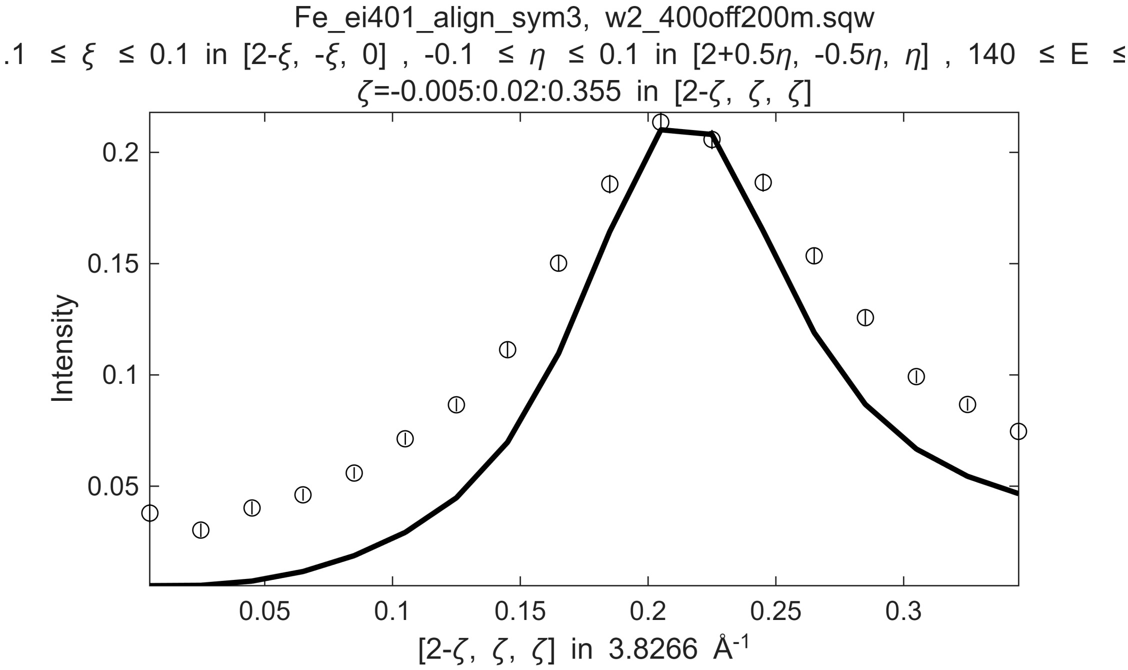

```
******************************
**** En = 140±10
******************************
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S= 1.1±0.06; J0= 35±0.42; gamma=52.9862±3.43407
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 130 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1


Fe_ei401_align_sym3, w2_400off200m.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 130 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

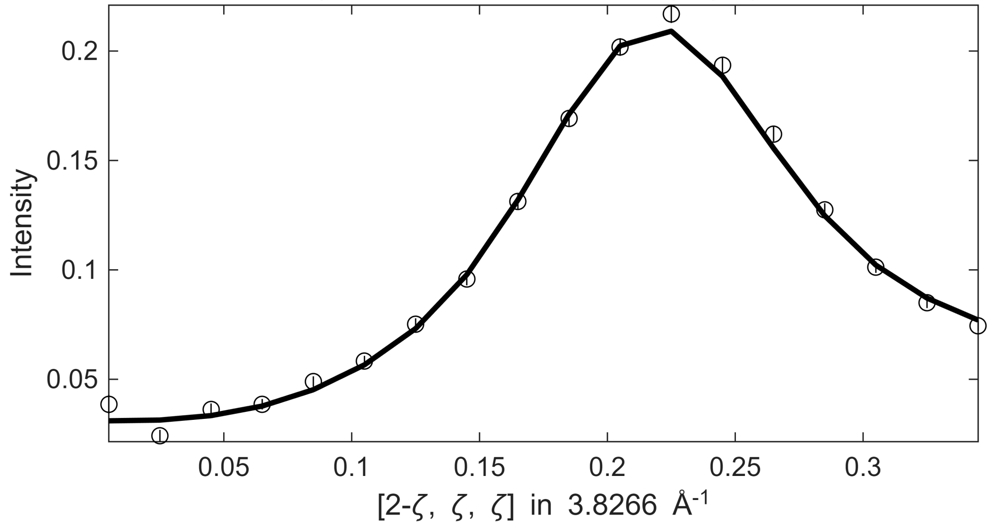
```
******************************
**** En = 150±10
******************************
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S= 1.1±0.042; J0= 35±0.22; gamma=51.0189±1.78744
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 140 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

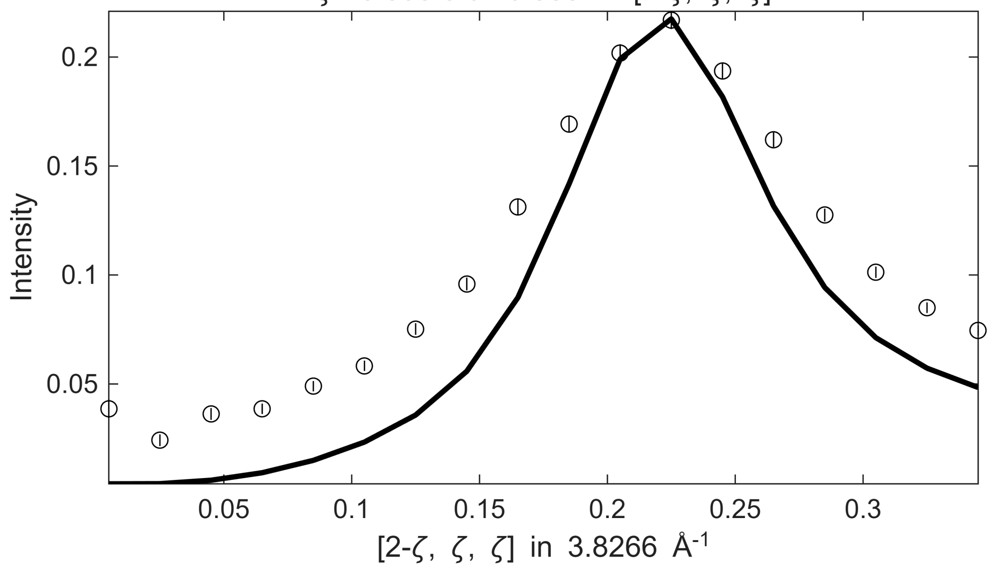

Fe_ei401_align_sym3, w2_400off200m.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 140 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

```
*****************************
**** En = 160±10
*****************************
```

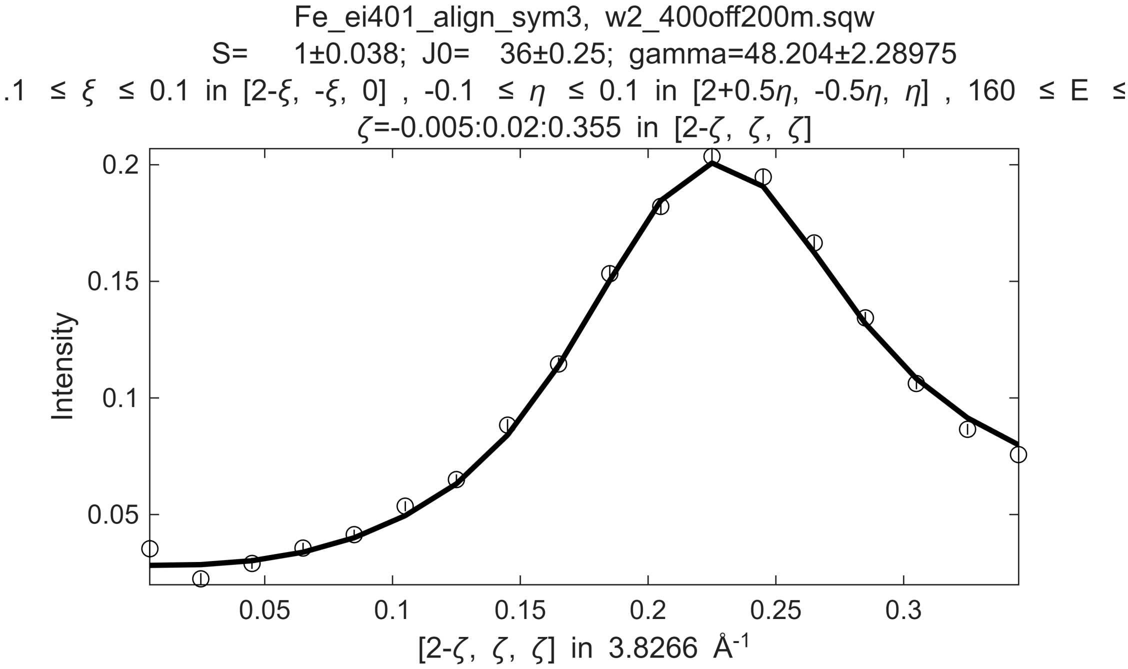
Fe_ei401_align_sym3, w2_400off200m.sqw
S= 1.1±0.04; J0= 36±0.26; gamma=47.8669±2.28238
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 150 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]

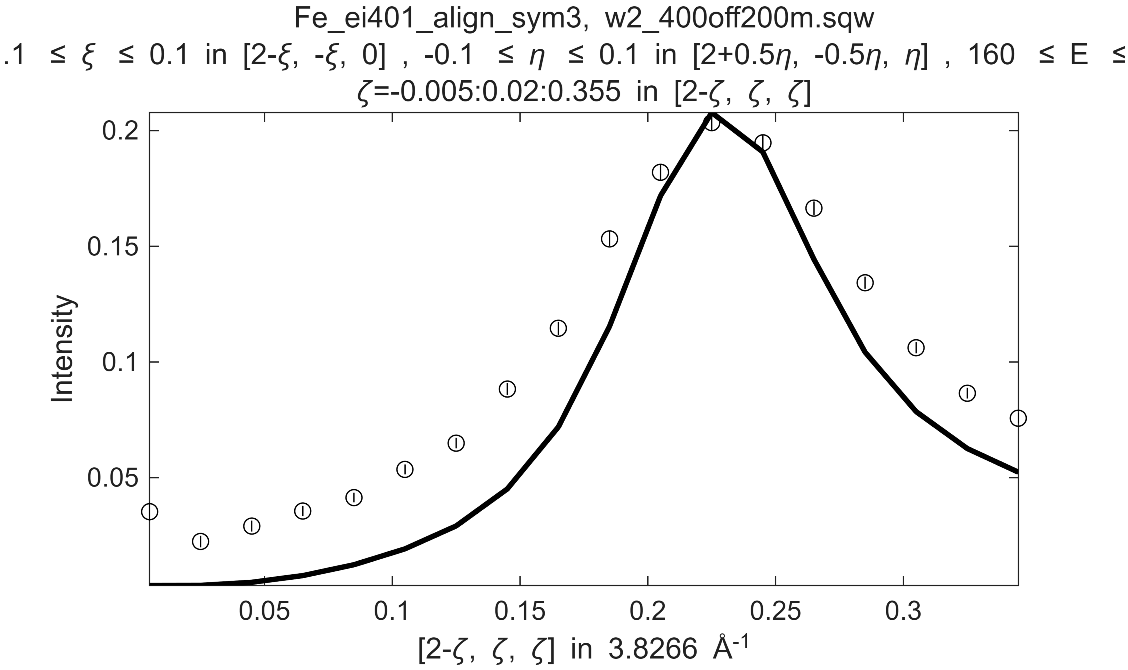


Fe_ei401_align_sym3, w2_400off200m.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 150 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]




```
******************************
**** En = 170±10
******************************
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S= 1±0.038; J0= 36±0.25; gamma=48.204±2.28975
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 160 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å^-1


Fe_ei401_align_sym3, w2_400off200m.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 160 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å^-1

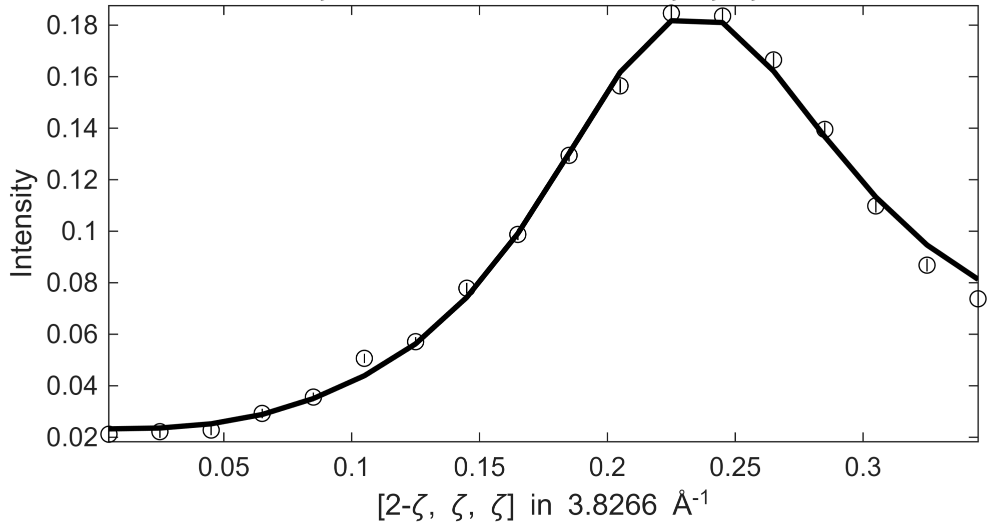
```
******************************
**** En = 180±10
******************************
```

Fe_ei401_align_sym3, w2_400off200m.sqw

S= 1.1±0.038; J0= 38±0.23; gamma=57.0996±2.27945

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 170 ≤ E ≤

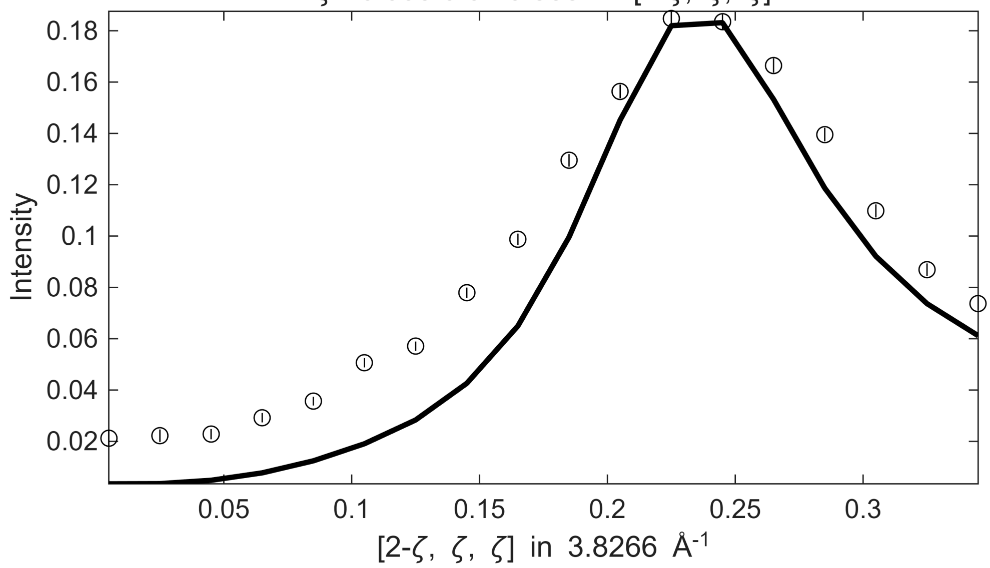
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]



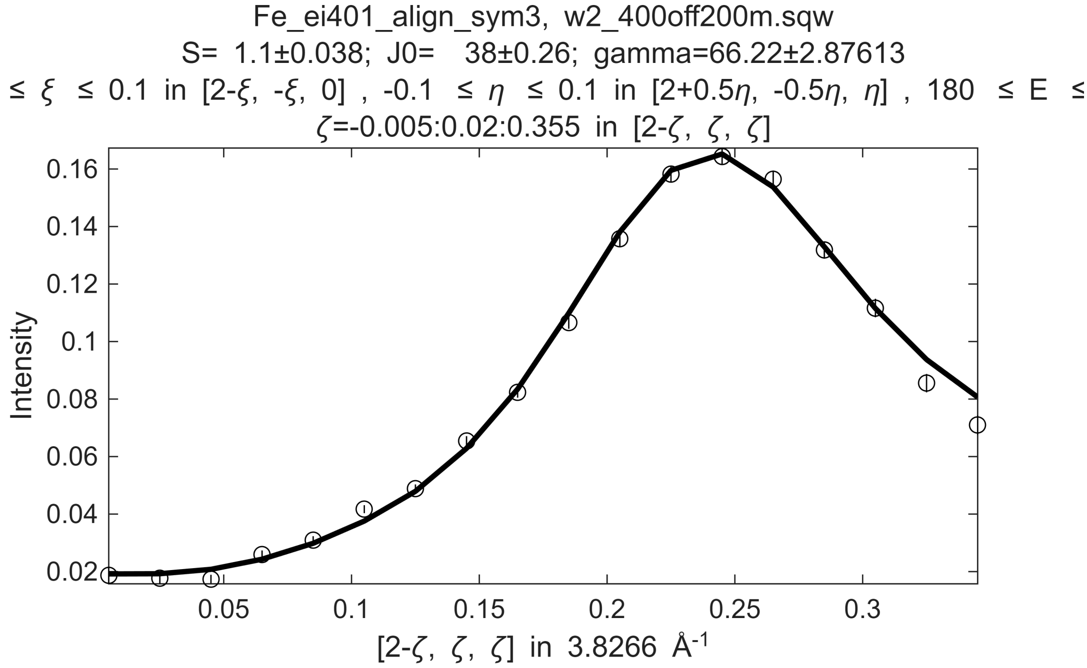
Fe_ei401_align_sym3, w2_400off200m.sqw

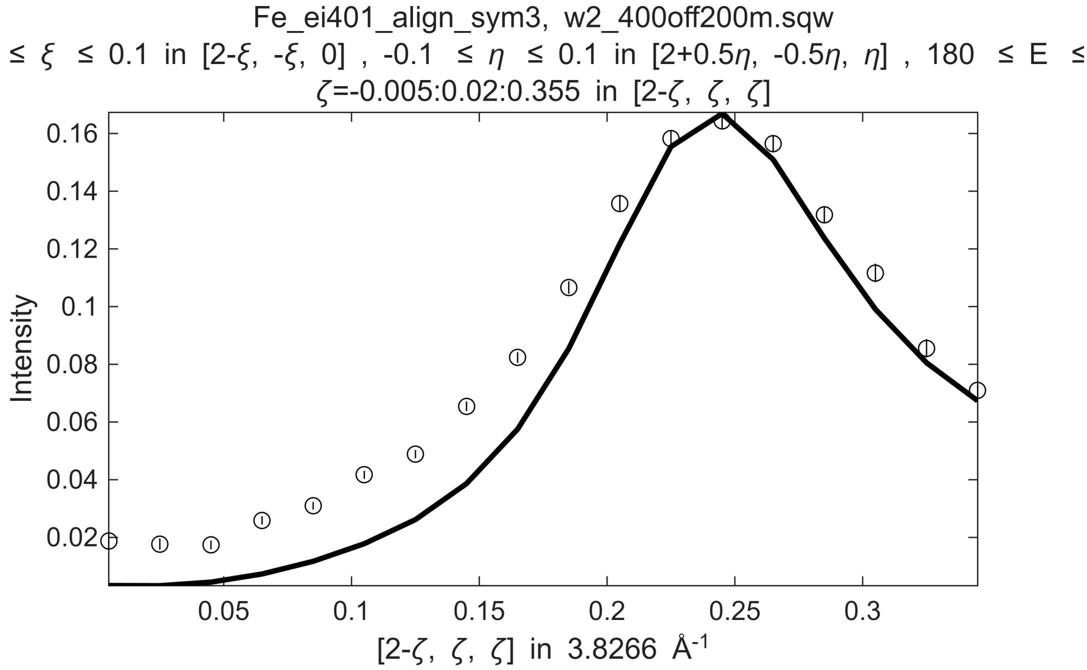
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 170 ≤ E ≤

ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]

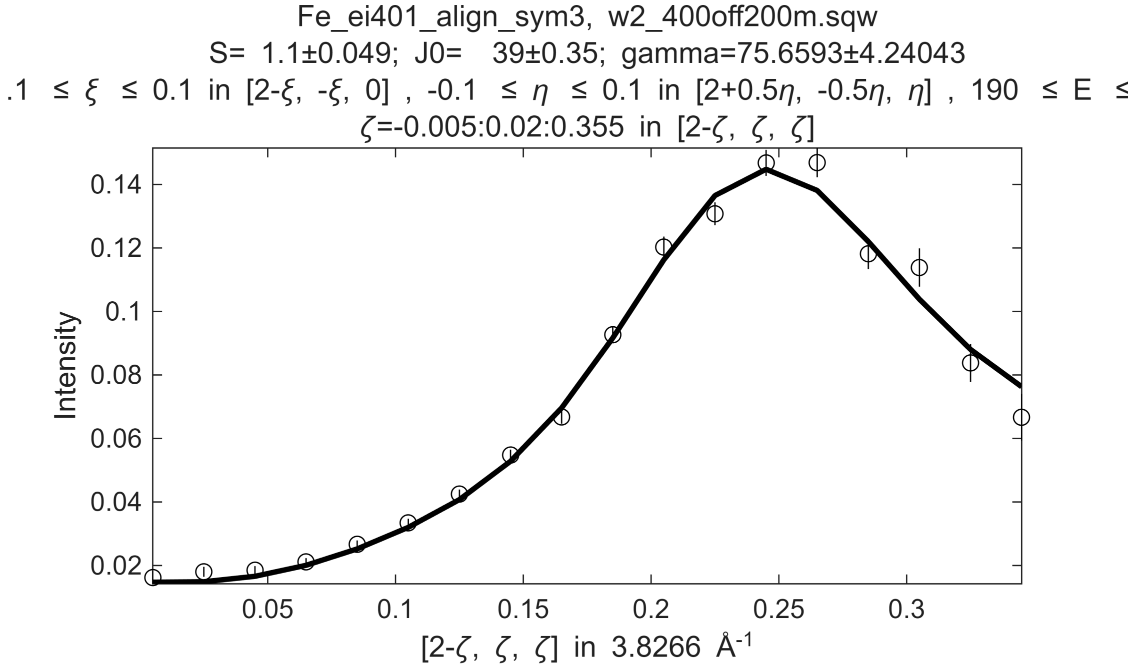


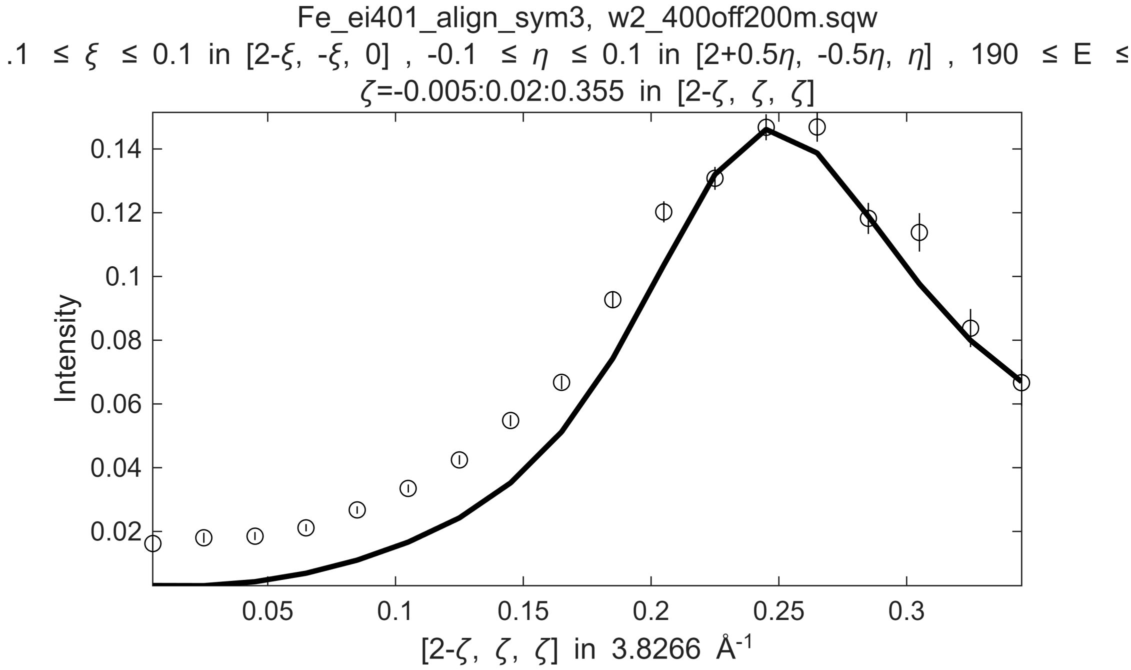

```
******************************
**** En = 190±10
******************************
```

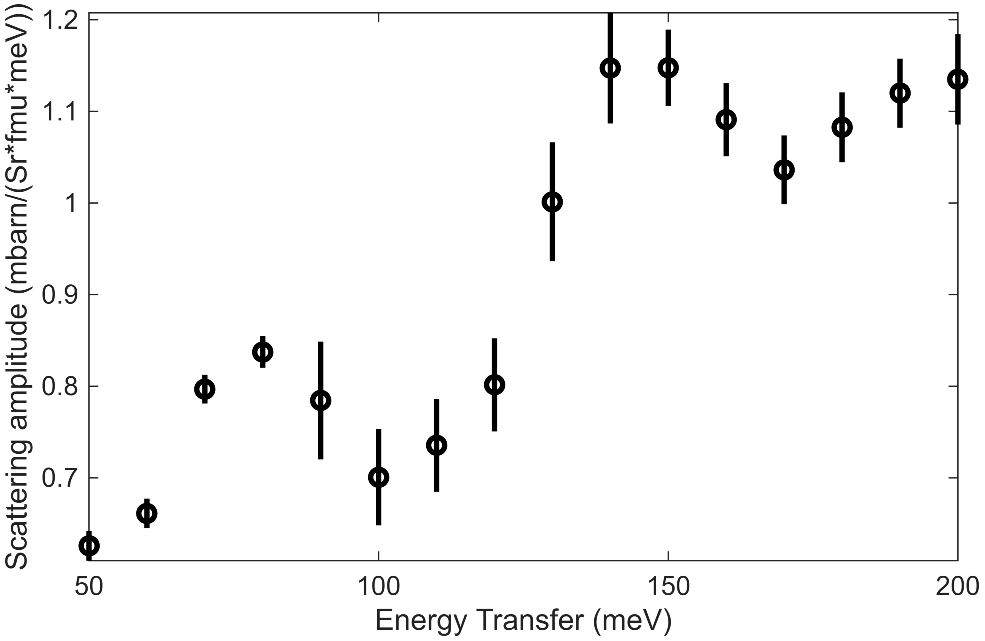


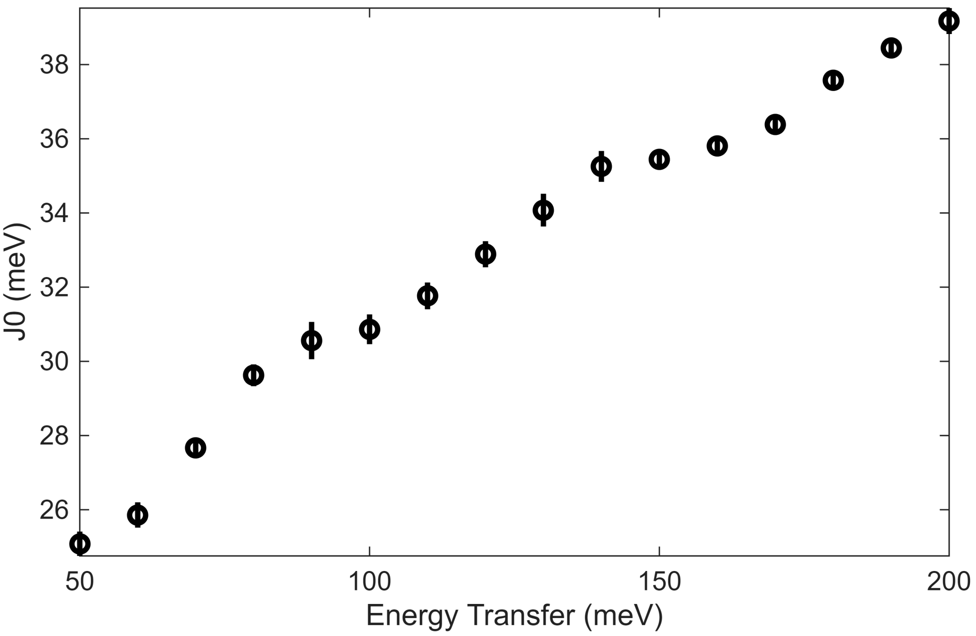


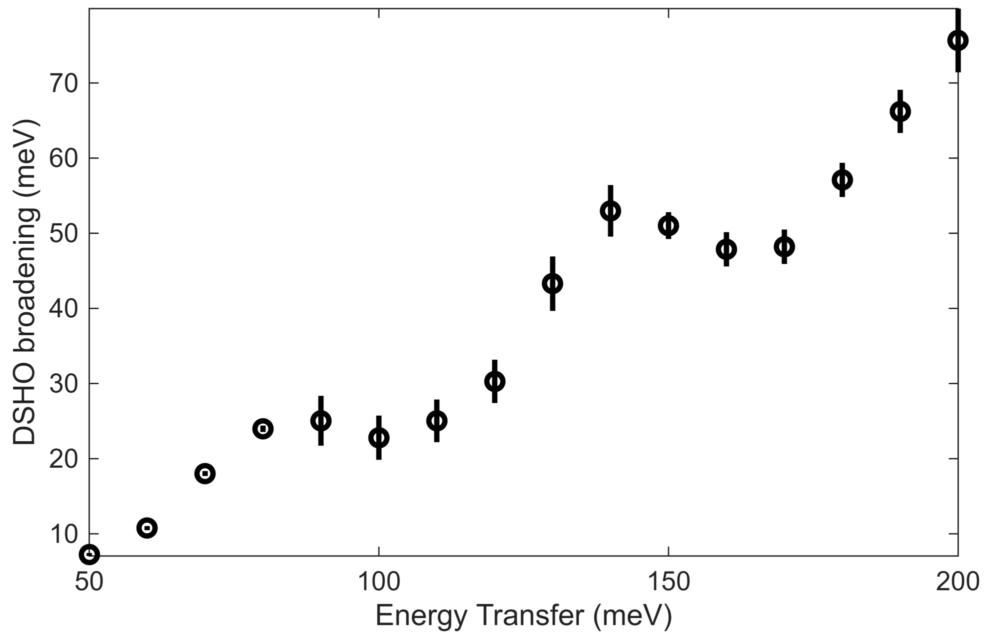

```
******************************
**** En = 200±10
******************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.92593
```


Fe_ei401_align_sym3, w2_400off200m.sqw
S= 1.1±0.049; J0= 39±0.35; gamma=75.6593±4.24043
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 190 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

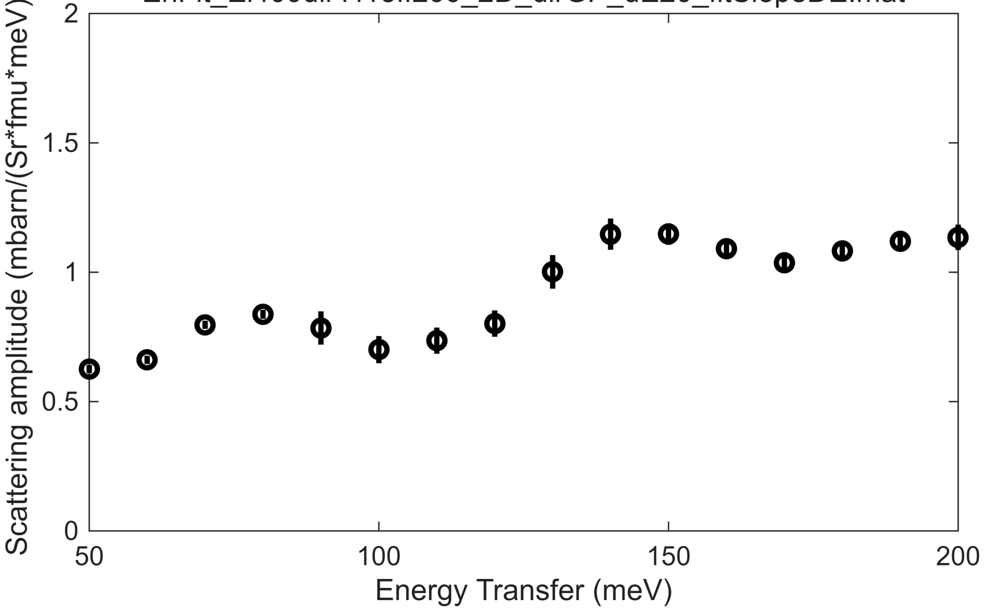

Fe_ei401_align_sym3, w2_400off200m.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 190 ≤ E ≤
ζ=-0.005:0.02:0.355 in [2-ζ, ζ, ζ]
Intensity
[2-ζ, ζ, ζ] in 3.8266 Å-1

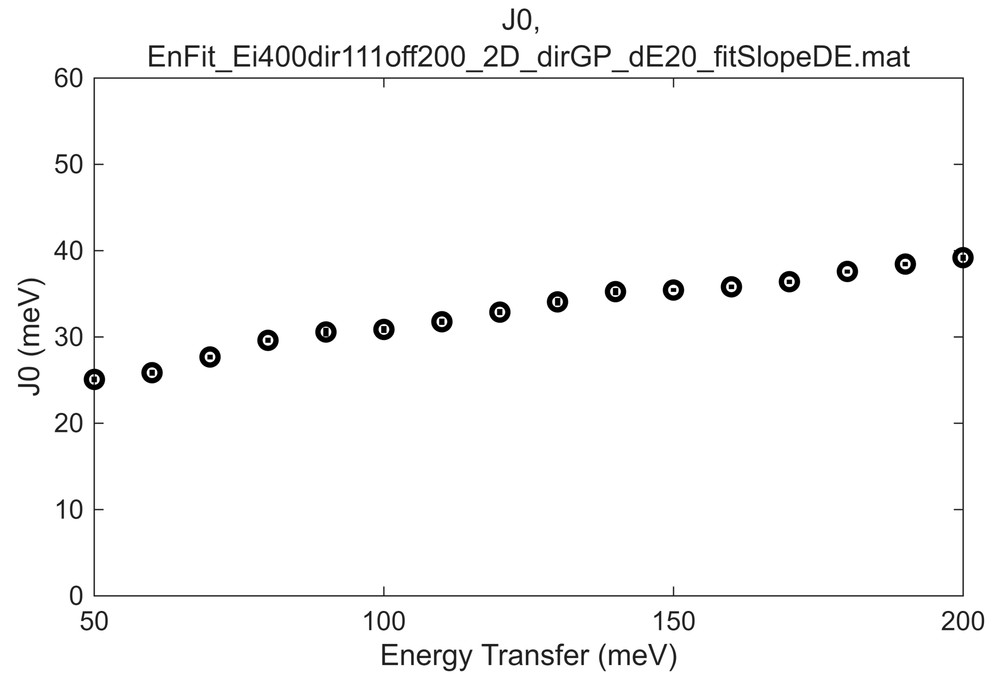

Scattering amplitude (mbarn/(Sr*fmu*meV))
1.2
1.1
1
0.9
0.8
0.7
50
100
150
200
Energy Transfer (meV)

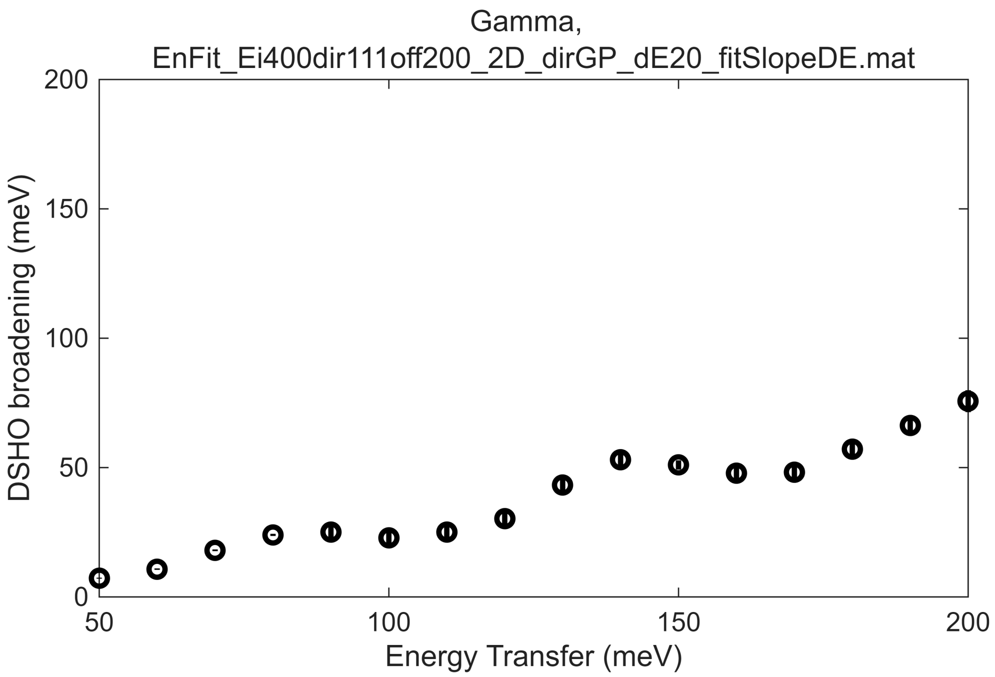

J0 (meV)
38
36
34
32
30
28
26
50
100
150
200
Energy Transfer (meV)



Amplitude,
EnFit_Ei400dir111off200_2D_dirGP_dE20_fitSlopeDE.mat

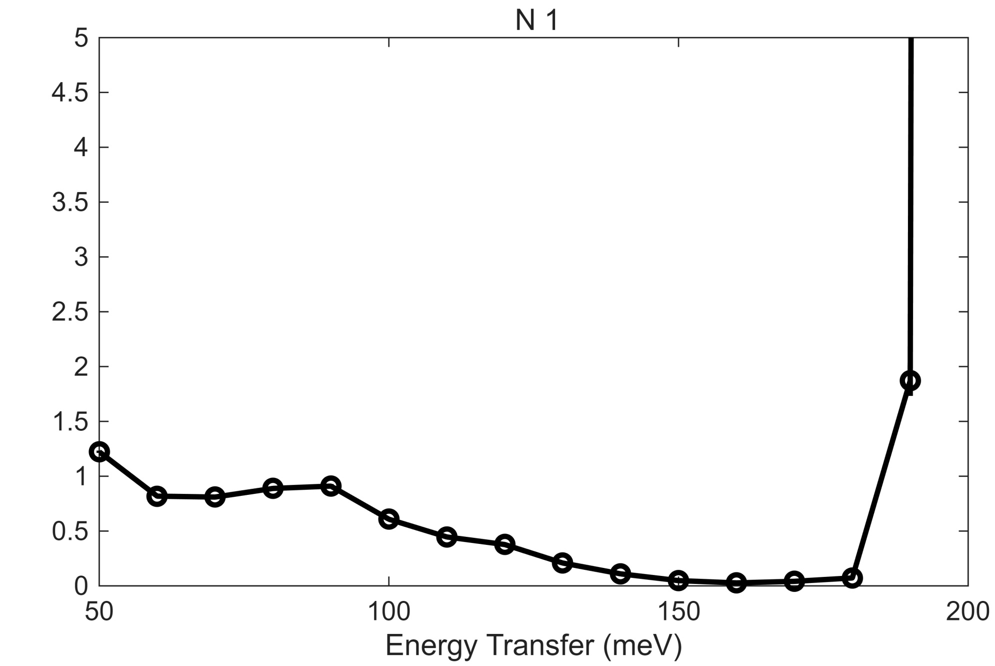




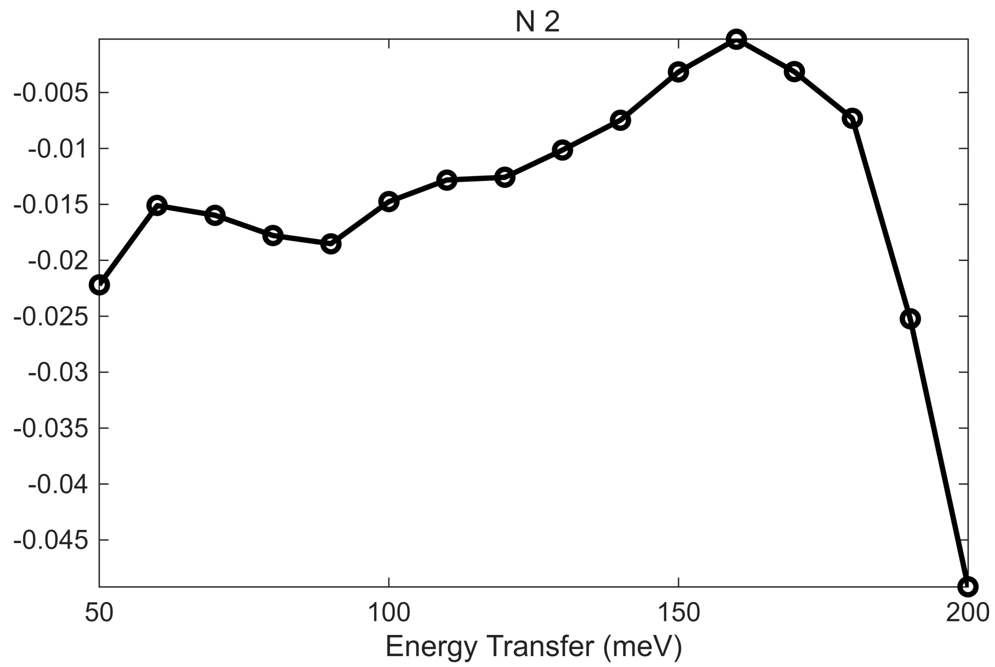



```
%cl,'En_cuts800',dir_name,cut_en,dE_step,half_dE);
beep
beep
beep
```

```
fit_res_400 = fit_res_400rec;

bg_data = extract_bg_par(fit_res_400.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
keep_figure; ly 0 5;
```




```
pd(bg_data(2));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
keep_figure
```




```
fit_resEi400 = fit_res_400rec;
save(res_name,"fit_resEi400")
```